

Supplementary material. Bat species used to test the relationship between apparent dry matter digestibility (D^*_m), climate, and mass-independent basal metabolic rate (BMR), of bats of the family Vespertilionidae. We measured BMR from 10 species of bats from the family Vespertilionidae and gathered information from 29 species from the bibliography ranging in body mass (M_b) from 3.4 g to 30.3 g. D^*_m was modeled using the formula: $D^*_m = -0.71 M_b + 86.45$, published by Cabrera-Campos et al. (2021). In this formula, M_b values were taken from bats measured previously to the metabolic rate estimations. For those bats we retrieved from the bibliography, we used the M_b reported in the publication.

Bat species	Locality	Climate	M_b (g)	BMR (mL O ₂ h ⁻¹)	Modeled D^*_m (%)	References
<i>Antrozous pallidus</i>	California, EE.UU	Cold	22.0	2.2	70.7	Licht and Leitner 1967
<i>Lasiurus cinereus</i>	PNLM, Mexico	Cold	17.7	2.6	73.8	Ayala-Berdon et al. 2025
			19.7	2.5	72.4	
	SCM, Mexico		23.5	2.5	69.7	
			21.8	2.5	70.9	
			20.9	2.6	71.5	
			23.6	2.6	69.6	
			24.8	2.5	68.7	
	Cold	18.0	2.6	73.6	This study	
		21.3	3.0	71.2		
		22.2	2.8	70.6		
		21.0	2.9	71.4		
	Xalapa Mexico	Warm	16.6	3.1	74.6	
15.8			3.4	75.1		
<i>Lasiurus intermedius</i>	Quilamula, Mexico	Warm	15.9	2.7	75.1	Ayala-Berdon and Medina-Bello 2024

	Florida, EE.UU		22.6	2.0	70.3	Genoud 1993
<i>Lasiurus ega</i>	La Mancha, Mexico	Warm	10.8	3.6	78.7	Ayala-Berdon et al. 2025
			10.8	5.7	78.7	
<i>Lasiurus borealis</i>	Florida, EE.UU	Warm	12.2	3.0	77.7	Genoud 1993
<i>Lasiurus frantzii</i>	SCM, Mexico	Cold	9.7	4.1	79.5	Ayala-Berdon et al. 2025
			10.2	3.1	79.1	
			9.2	2.6	88.5	
	Autlán, Mexico	Warm	8.4	5.8	80.4	This study
	La Mancha, Mexico		9.2	3.4	79.8	
<i>Lasiurus seminolus</i>	EE.UU	Warm	9.7	2.7	79.5	Genoud 1990
			9.0	2.2	80.0	
	Florida, EE.UU	Warm	9.8	2.7	79.4	Genoud 1993
			9.0	2.2	80.0	
<i>Eptesicus serotinus</i>	-	Cold	27.0	2.4	67.2	Speakman et al. 2003
<i>Eptesicus fuscus</i>	PNLM, Mexico	Cold	14.3	2.4	76.2	Ayala-Berdon et al. 2025
			14.2	2.6	76.1	
			16.5	2.5	74.6	
			15.7	2.5	75.2	
	SCM, Mexico		17.6	2.6	73.9	
			11.2	2.7	78.4	
			15.4	3.3	75.4	
			13.8	2.8	77.0	
	Autlán, Mexico	Warm	14.9	4.2	75.8	This study

	SCM, Mexico		15.1	3.7	75.6	
			15.6	4.8	75.3	
			15.4	3.3	75.4	
			13.8	2.8	76.6	
			14.2	2.4	76.3	
			14.5	4.0	76.1	
<i>Eptesicus brasiliensis</i>	Sarapiquí, Costa Rica	Warm	8.0	3.9	80.7	Ayala-Berdon et al. unpublished data
			9.0	4.1	80.0	
<i>Eptesicus furinalis</i>	El Limón, Mexico	Warm	6.2	7.3	82.0	Ayala-Berdon and Medina-Bello 2024
	La Mancha, Mexico	Warm	8.1	5.5	80.6	
			8.4	5.4	80.4	
			8.8	6.8	80.1	
	Saraquí, Costa Rica	Warm	9.0	4.7	80.0	Ayala-Berdon et al. unpublished data
			9.0	4.1	80.0	
<i>Nycticeius humeralis</i>	EE.UU Florida, EE.UU	Warm	9.0	2.3	80.0	Genoud 1990
			11.0	1.6	78.6	Genoud 1993
			9.0	2.3	80.0	
			11.0	1.8	78.5	
<i>Nyctophilus gouldi</i>	New South Wales, Australia	Warm	10.0	2.5	79.3	Geiser and Brigham 2000
	New South Wales, Australia	Warm	9.9	2.4	79.3	Currie et al. 2015
<i>Nyctalus geoffroyi</i>	Tasmania, Australia	Warm	9.8	2.2	79.4	Dixon and Rose 2003

	New South Wales, Australia	Warm	7.0	2.4	81.4	Geiser and Brigham 2000
	Perth, Australia	Warm	8.0	2.5	80.7	Hosken and Withers 1999
<i>Plecotus auritus</i>	Germany	Cold	7.4	2.2	80.8	Sørås et al. 2022
<i>Pipistrellus pipitrellus</i>	EE.UU	Cold	6.0	3.6	82.1	Speakman et al. 1989
<i>Pipistrellus kuhlii</i>	Negev, Israel	Cold	6.9	2.4	81.5	Muñoz-García et al. 2012
<i>Rhogeessa parvula</i>	El Limón, Mexico	Warm	4.3	10.7	83.3	Ayala-Berdon and Medina-Bello 2024
			4.1	9.0	83.5	
			4.2	10.9	83.4	
<i>Rhogeessa alleni</i>	SCM, Mexico		3.7	11.8	83.9	Ayala-Berdon et al. 2025
			3.7	11.6	83.9	
			3.8	9.3	83.7	
			3.5	10.7	83.9	
			3.6	11.3	83.8	
			3.4	10.4	84.0	
<i>Rhogeessa io</i>	Sarapiquí, Costa Rica		6.0	5.4	82.1	Ayala-Berdon et al. unpublished data
			5.0	4.9	82.8	
			5.0	4.8	82.8	
			5.0	5.7	82.8	
			5.0	5.2	82.8	
			6.0	4.8	82.1	

<i>Rhogeessa tumida</i>	La Mancha, Mexico		3.8	4.8	83.7	This study
			4.0	4.7	83.6	
			4.4	5.5	83.3	
			4.8	5.5	83.0	
	Xalapa, Mexico		3.5	10.7	83.9	
			3.6	11.2	83.8	
			3.4	10.4	84.0	
<i>Myotis myotis</i>	EE.UU	Cold	25.0	4.2	68.6	Hanus 1959
<i>Myotis velifer</i>	Quilamula, Mexico	Warm	8.6	3.8	80.3	Ayala-Berdon et al. unpublished data
			9.2	3.9	79.8	
			9.1	3.1	79.9	
			10.4	2.9	79.0	
			10.2	2.9	79.1	
			9.9	3.1	79.3	
			9.4	4.0	79.7	
			9.3	3.0	79.8	
			9.7	3.1	79.5	
			10.2	3.0	79.1	
			8.8	3.5	80.1	
	Quilamula, Mexico	Warm	8.9	4.1	80.1	This study
			9.2	3.8	79.8	
			9.4	3.9	79.7	
			9.9	4.4	79.3	
	Xalapa, Mexico	Warm	9.6	3.9	79.6	
			7.8	2.7	80.8	
<i>Myotis bechsteinii</i>	-	Germany	8.9	2.0	80.1	Becker et al. 2013

<i>Myotis lucifugus</i>	Boston, EE.UU	Cold	7.6	4.4	80.9	Kurta and Kunz 1988
	SCM, Mexico	Cold	7.6	3.6	81.0	This study
<i>Myotis austroriparius</i>	Florida, EE.UU	Warm	7.5	2.1	81.1	McNab 1969
<i>Myotis daubentonii</i>	Germany	Cold	7.8	1.8	80.8	Becker et al. 2013
<i>Myotis auriculus</i>	Hueyotlipan, Mexico		6.9	4.0	81.5	Ayala-Berdon et al. 2025
<i>Myotis thysanodes</i>	SCM, Mexico	Cold	6.6	6.3	81.7	This study
			6.4	5.8	81.8	
	SCM, Mexico		5.6		82.4	
				4.2		
<i>Myotis volans</i>	PNLM, Mexico	Cold	5.1	3.2	82.8	Ayala-Berdon et al. 2025
			5.5	2.8	82.5	
	SCM, Mexico	Cold	5.9	3.3	82.2	
			5.7	3.6	82.3	
			6.3	4.4	81.9	
			5.0	5.1	82.8	
			6.2	4.5	82.0	
			5.3	4.1	82.6	
			6.4	3.8	81.8	
	SCM, Mexico	Cold	5.4	5.1	82.6	This study
			5.8	5.5	82.3	
			5.4	5.6	82.6	
			4.8	4.7	83.0	

<i>Myotis albescens</i>	Sarpiquí, Costa Rica	Warm	5.5	5.1	82.5	Ayala-Berdon et al. unpublished data
<i>Myotis elegans</i>	Sarpiquí, Costa Rica	Warm	5.0	4.4	82.8	Ayala-Berdon et al. unpublished data
<i>Myotis keaysi</i>	Prusia, Costa Rica	Warm	6.5	2.9	81.8	Ayala-Berdon et al. unpublished data
			7.0	2.5	81.4	
	La Mancha, Mexico	Warm	3.8	3.3	83.7	Ayala-Berdon et al. 2025
			4.0	3.9	83.6	
			4.0	5.2	83.6	
<i>Myotis fortidens</i>	La Mancha, Mexico	Warm	5.6	5.3	82.4	Ayala-Berdon et al. 2025
	Autlán, Mexico		5.1	3.3	4.4	
	La Mancha, Mexico	Warm	4.7	4.4	83.1	This study
<i>Myotis californicus</i>	PNLM, Mexico	Cold	3.9	5.2	83.6	Ayala-Berdon and Medina-Bello 2024
			4.8	4.7	83.0	
	SCM, Mexico	Cold	4.4	5.6	83.3	Ayala-Berdon et al. 2025
			3.9	5.2	83.6	
			4.4	4.9	83.3	
	SCM, Mexico	Cold	3.8	4.4	83.7	This study
			4.2	4.5	83.4	
<i>Myotis melanorhinus</i>	Hueyotlipan, Mexico	Cold	4.3	3.8	83.3	Ayala-Berdon et al. 2025
	SCM, Mexico	Cold	4.6	4.0	83.1	
			4.6	5.4	83.1	

<i>Myotis nigricans</i>	Florida, EE.UU Sarapiquí, Costa Rica	Warm	3.7	1.9	83.8	McNab 1989
			6.0	4.0	82.1	Ayala-Berdon et al. unpublished data
			6.0	3.5	82.1	
			5.0	4.3	82.8	
			6.0	3.9	82.1	
			6.0	3.8	82.1	
<i>Myotis mystacinus</i>	Nittedal, Norway	Cold	4.4	2.3	83.3	Skåra et al. 2021
<i>Miniopterus schreibersii</i>	Darwin, Australia	Warm	11.0	4.8	78.6	Baudinette et al. 2000

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Thermoneutral zone (*TNZ*) of bat species from the family Vespertilionidae. The table presents the lower (*T_{LC}*) and upper (*T_{UC}*) limits of the thermoneutral zone for each species, determined under controlled laboratory conditions. These values were used as reference points to estimate the basal metabolic rate (*BMR*) of each species in this study.

Species	<i>TNZ</i> (°C)		References
	<i>T_{LC}</i> (°C)	<i>T_{UC}</i> (°C)	
<i>Lasiurus cinereus</i>	21.2	33.4	Ayala-Berdon and Medina Bello 2024
<i>Lasiurus ega</i>	25.0	33.0	Ayala-Berdon et al. 2025
<i>Lasiurus frantzii</i>	24.0	22.6	Ayala-Berdon and Medina Bello 2024
<i>Eptesicus fuscus</i>	21.2	34.6	Ayala-Berdon and Medina Bello 2024
<i>Eptesicus furinalis</i>	31.0	38.0	Ayala-Berdon et al. 2025
<i>Rhogeessa tumida</i>	31.5	37.5	Ayala-Berdon et al. 2025
<i>Myotis velifer</i>	27.5	35.3	Ayala-Berdon and Medina Bello 2024
<i>Myotis fortidens</i>	27.0	35.5	Ayala-Berdon et al. 2025

<i>Myotis volans</i>	28.7	37.5	Ayala-Berdon and Medina Bello 2024
<i>Myotis californicus</i>	27.5	37.5	Ayala-Berdon and Medina Bello 2024

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